

Batter Overall Score and Park Factor Analysis

Exploratory Research Methodology

1. Overall Score: Player Skill Profile

Overall Score measures a player's statistical skill profile using batting average, on-base percentage, slugging percentage, strikeout rate, and walk rate.

Overall Score = (0.25 x BA Score) + (0.25 x OBP Score) + (0.25 x SLG Score) + (0.15 x K% Score) + (0.10 x BB% Score)

BA, OBP, and SLG each receive 25% weight, K% receives 15%, and BB% receives 10%. Each component score is capped between 0 and 10.

2. Park Factors

Park factors estimate how a ballpark influences offensive outcomes relative to a neutral environment. This analysis uses 2024-2026 rolling Statcast park factors, where 100 represents a neutral park environment.

The park-factor dataset includes separate factors for hits, singles, doubles, triples, home runs, walks, strikeouts, on-base percentage, and other offensive outcomes.

3. Park-Adjusted Overall

A challenger version of Overall Score was constructed to isolate the effect of ballpark environment without changing the underlying Batter Analyzer scoring model.

BA, OBP, and SLG were reconstructed using the relevant park-adjusted batting components. K% and BB% scoring remained unchanged.

Because a player does not play his entire season in his home ballpark, seasonal park exposure was approximated as 50% home park and 50% neutral environment.

PF Effective = 0.5 x (PF / 100) + 0.5

Adjusted Component = Observed Component / PF Effective

For example, a park factor of 117 produces an effective season factor of 1.085 under the 50% home / 50% neutral approximation.

The adjusted BA, OBP, and SLG values were then passed through the original Batter Analyzer scoring functions. No Overall Score weights, thresholds, clamps, or scoring ranges were changed.

4. Overall Change

Overall Change measures the difference between the park-adjusted challenger and the original Batter Analyzer Overall Score.

Overall Change = Park-Adjusted Overall - Original Overall

Positive change means park adjustment increased Overall; negative change means park adjustment decreased Overall; zero means the two Overall Scores are equal.

5. Research Results

The analysis included 433 2026 hitters with more than 50 at-bats who could be cleanly matched to one of the 29 ballparks contained in the supplied park-factor dataset.

- Median absolute Overall change: 0.125
- Mean absolute Overall change: 0.215
- 74.6% of players changed by no more than +/-0.30 Overall points
- 91.0% of players changed by no more than +/-0.50 Overall points
- 97.0% of players changed by no more than +/-1.00 Overall points
- Pearson correlation: 0.984
- Spearman rank correlation: 0.987
- Mean directional change: +0.012

The largest changes were concentrated among players associated with more extreme park environments. Colorado hitters accounted for several of the largest downward adjustments, while several substantial upward adjustments occurred among hitters associated with pitcher-friendly environments such as Seattle.

6. Research Objective

The purpose of this analysis is to test whether ballpark environment materially changes Batter Analyzer Overall Scores or player ordering when the original five-metric scoring architecture is preserved.

The primary outcomes being examined are:

- Absolute change in Overall Score
- Percentage of players remaining within defined Overall-change ranges
- Correlation between original and park-adjusted Overall Scores
- Correlation between original and park-adjusted player rankings

Research Question

Does adjusting for ballpark environment materially change Batter Analyzer Overall Scores or player rankings?

Interpretation

This is an exploratory analysis. The results do not demonstrate that park factors are irrelevant. Within this sample, however, park adjustment produced relatively small changes for most players and very little change in overall player ordering.

Larger movements were concentrated in more extreme park environments, suggesting that park context may be more consequential at the edges of the park-factor distribution than across the player population as a whole.

The analysis approximates seasonal exposure as 50% home park and 50% neutral rather than reconstructing each player's actual plate appearances by ballpark. Future analysis can repeat the test across additional seasons and use more precise park-by-park exposure to determine whether the observed model stability persists.